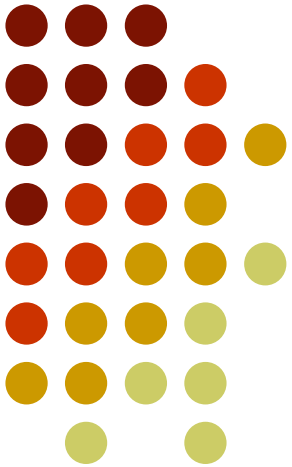
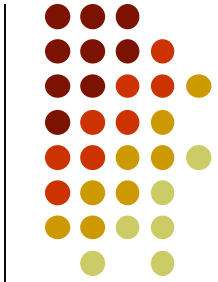
Mechanics of Futures Markets

Chapter 2

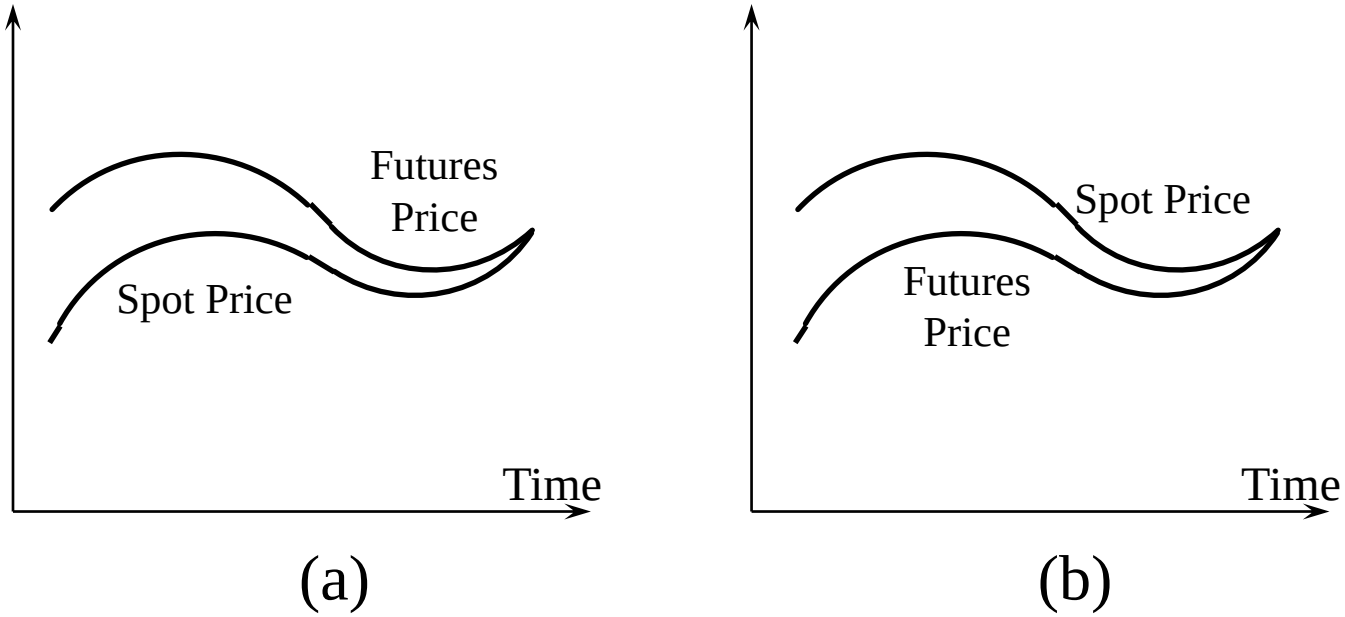


Futures Contracts



- Available on a wide range of underlyings
- Exchange traded
- Specifications need to be defined:
  - What can be delivered,
  - Where it can be delivered, &
  - When it can be delivered
- Settled daily

Convergence of Futures to Spot (Figure 2.1, page 28)



Margin

- A margin is cash or marketable securities deposited by an investor with his or her broker
  - Initial margin
  - Maintenance margin
  - Variation margin
- The balance in the margin account is adjusted to reflect daily settlement
- The investor is entitled to withdraw any balance in the margin account in excess of the initial margin
- Margin minimizes the possibility of a loss through a default on a contract

Example of a Futures Trade (page 29-30)



- An investor takes a long position in 2 December gold futures contracts on June 5
  - contract size is 100 oz.
  - futures price is US$1250
  - initial margin requirement is US$6,000/contract (US$12,000 in total)
  - maintenance margin is US$4,500/contract (US$9,000 in total)

A Possible Outcome (Table 2.1, page 30)

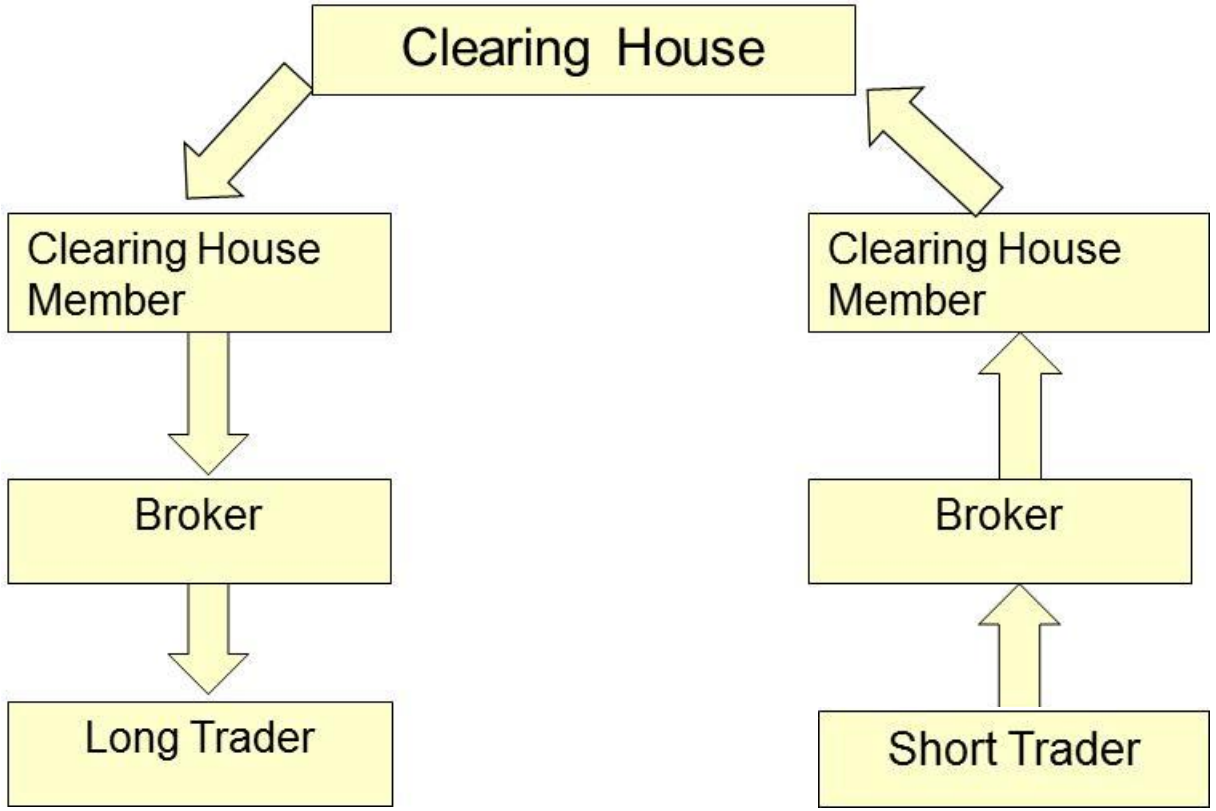
Day	*Trade price ($)*	*Settlement price ($)*	*Daily gain ($)*	*Cumulative gain ($)*	*Margin account balance ($)*	*Margin call ($)*
1	1,250.00				12,000	
1		1,241.00	−1,800	−1,800	10,200	
2		1,238.30	−540	−2,340	9,660	
3		1,244.60	1,260	−1,080	10,920	
4		1,241.30	−660	−1,740	10,260	
5		1,240.10	−240	−1,980	10,020	
6		1,236.20	−780	−2,760	9,240	
7		1,229.90	−1,260	−4,020	7,980	4,020
8		1,230.80	180	−3,840	12,180	
9		1,225.40	−1,080	−4,920	11,100	
10		1,228.10	540	−4,380	11,640	
11		1,211.00	−3,420	−7,800	8,220	3,780
12		1,211.00	0	−7,800	12,000	
13		1,214.30	660	−7,140	12,660	
14		1,216.10	360	−6,780	13,020	
15		1,223.00	1,380	−5,400	14,400	
16	1,226.90		780	−4,620	15,180	

Key Points About Futures

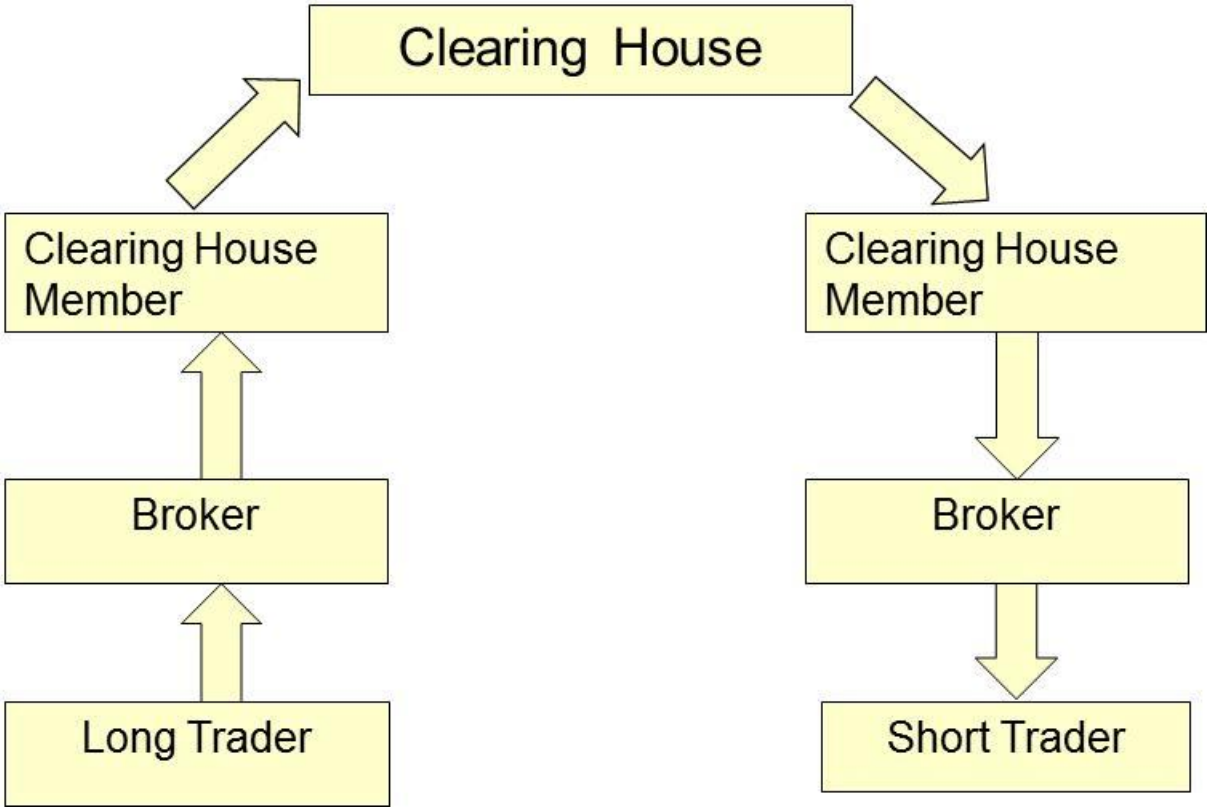


- They are settled daily
- Closing out a futures position involves entering into an offsetting trade
- Most contracts are closed out before maturity

Margin Cash Flows When Futures Price Increases



Margin Cash Flows When Futures Price Decreases



Collateralization in OTC Markets



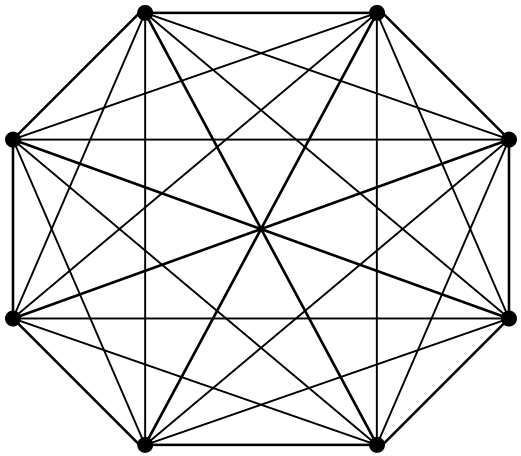
- It is becoming increasingly common for transactions to be collateralized in OTC markets
- Consider transactions between companies A and B
- The credit support annex (CSA) might require A to post collateral with B equal to the value to B of its outstanding transactions with B when this value is positive.

Clearing Houses and OTC Markets

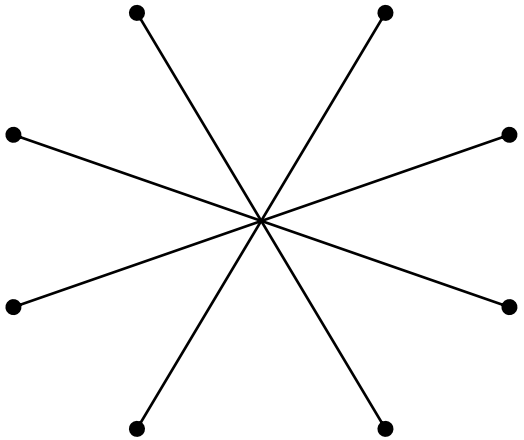


- Traditionally transactions have been cleared bilaterally in OTC markets
- Following the 2007-2009 crisis, the has been a requirement for most standardized OTC derivatives transactions to be cleared centrally though clearing houses.

Bilateral Clearing vs Central Clearing House



Bilateral clearing



Central clearing

Some Terminology



- Open interest: the total number of contracts outstanding
  - equal to number of long positions or number of short positions
- Settlement price: the price just before the final bell each day
  - used for the daily settlement process
- Volume of trading: the number of trades in one day

Crude Oil Trading on July 13, 2012 (from Table 2.2, page 35)



	Open	High	Low	Prior settle	Last trade	Change	Volume
Aug 2012	85.86	87.61	85.58	86.08	87.28	+1.20	223,698
Sept 2012	86.33	88.00	85.95	86.46	87.68	+1.22	87,931
Dec 2012	87.45	89.21	87.39	87.73	88.94	+1.21	31,701
Dec 2013	88.85	90.15	88.78	88.92	89.95	+1.03	11,128
Dec 2014	87.20	87.74	87.20	86.98	87.74	+0.76	2,388

Questions



- When a new trade is completed what are the possible effects on the open interest?
- Can the volume of trading in a day be greater than the open interest?

Delivery

- Very few of the futures contracts that are entered into lead to delivery of the underlying asset. Most are closed out early.
- If a futures contract is not closed out before maturity, it is usually settled by delivering the assets underlying the contract. When there are alternatives about what is delivered, where it is delivered, and when it is delivered, the party with the short position chooses.
- A few contracts (for example, those on stock indices and Eurodollars) are settled in cash.

Types of Orders



- Market order
  - a request that a trade be carried out immediately at the best price available in the market
- Limit order
  - specifies a particular price
- Stop-loss order
  - specifies a particular price. The order is executed at the best available price once a bid or offer is made at that price or a *less-favorable* price.
- Stop-limit order
  - a combination of a stop order and a limit order

Types of Orders (con't)

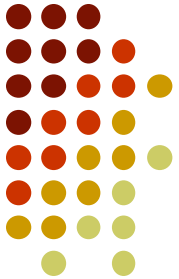


- Market-if-touched (MIT) order
  - Executed at the best available price after a trade occurs at the price or a *more favorable* price
- Discretionary order
  - Traded as a market order except that execution may be delayed at the broker's discretion in an attempt to get a better price.
- Time-of-day order
  - Specifies a particular period of time during the day when the order can be executed

Types of Orders (con't)

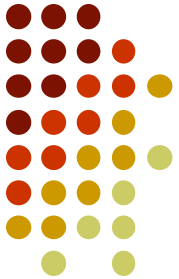
- Open order
  - Be in effect until executed or until the end of trading in the particular contract
- Fill-or-kill order
  - Be executed immediately on receipt or not at all

Forward Contracts



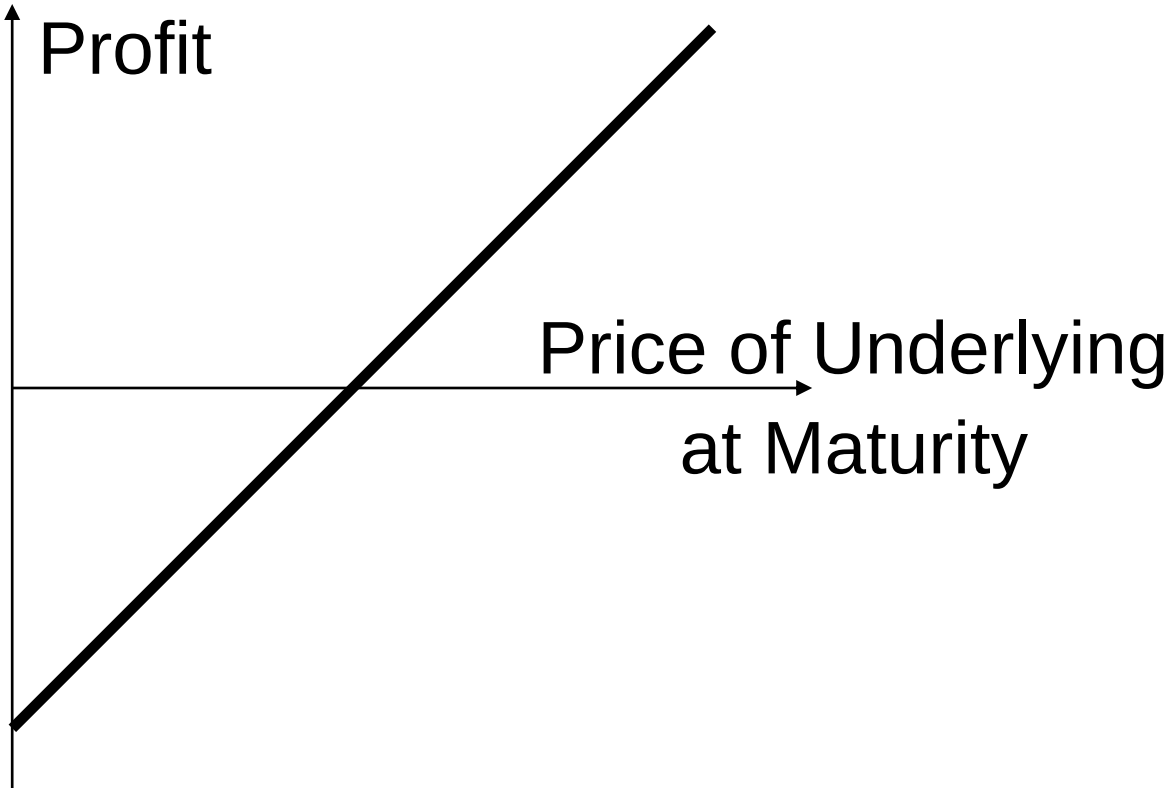
- A forward contract is an OTC agreement to buy or sell an asset at a certain time in the future for a certain price
- There is no daily settlement (but collateral may have to be posted). At the end of the life of the contract one party buys the asset for the agreed price from the other party

Forward Contracts vs Futures Contracts (Table 2.3, page 43)

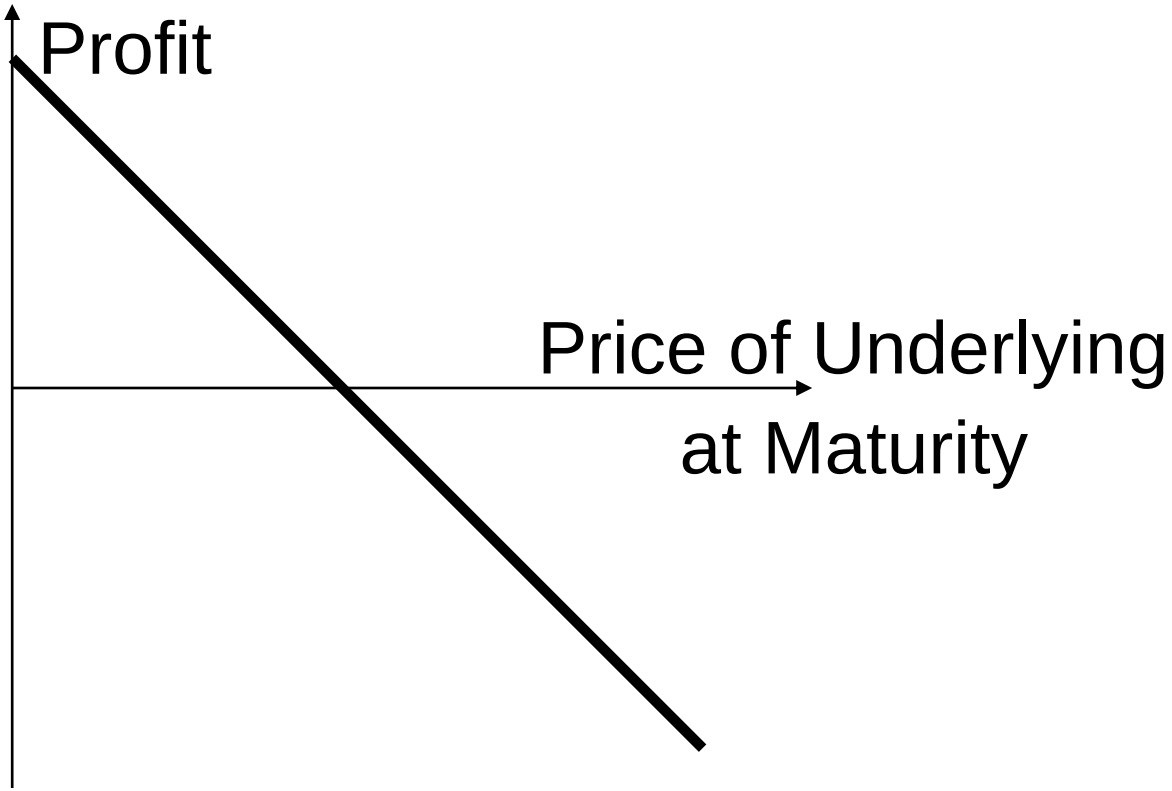


Forward	*Futures*
Private contract between two parties	Traded on an exchange
Not standardized	Standardized
Usually one specified delivery date	Range of delivery dates
Settled at end of contract	Settled daily
Delivery or final settlement usual	Usually closed out prior to maturity
Some credit risk	Virtually no credit risk

Profit from a Long Forward or Futures Position



Profit from a Short Forward or Futures Position



Foreign Exchange Quotes



- Futures exchange rates are quoted as the number of USD per unit of the foreign currency
- Forward exchange rates are quoted in the same way as spot exchange rates. This means that GBP, EUR, AUD, and NZD are USD per unit of foreign currency. Other currencies (e.g., CAD and JPY) are quoted as units of the foreign currency per USD.

Summary

- Futures price will converge to spot price as $t \rightarrow$ maturity.
- Margin:
  - Initial margin, maintenance margin, variation margin
  - Daily settlement practice
- Open interest
  - Total number of contracts outstanding = number of long/short positions

Summary

- Main types of orders
  - Market order, limit order, stop-loss order, stop-limit order
- Forward contracts vs Futures contracts

Forward	***Futures***
Private contract between two parties	Traded on an exchange
Not standardized	Standardized
Usually one specified delivery date	Range of delivery dates
Settled at end of contract	Settled daily
Delivery or final settlement usual	Usually closed out prior to maturity
Some credit risk	Virtually no credit risk